Bayesian Classification: Why?

- A statistical classifier: performs *probabilistic prediction, i.e.,* predicts class membership probabilities
- Foundation: Based on Bayes' Theorem.
- Performance: A simple Bayesian classifier, *naïve Bayesian classifier*, has comparable performance with decision tree and selected neural network classifiers
- Incremental: Each training example can incrementally increase/decrease the probability that a hypothesis is correct — prior knowledge can be combined with observed data
- Standard: Even when Bayesian methods are computationally intractable, they can provide a standard of optimal decision making against which other methods can be measured

Bayes' Theorem: Basics

- Bayes' Theorem:

$$P(H|\mathbf{X}) = \frac{P(\mathbf{X}|H)P(H)}{P(\mathbf{X})} = P(\mathbf{X}|H) \times P(H) / P(\mathbf{X})$$

  - Let **X** be a data sample ("*evidence*"): class label is unknown
  - Let H be a *hypothesis* that X belongs to class C
  - Classification is to determine P(H|**X**), (i.e., *posteriori probability):* the probability that the hypothesis holds given the observed data sample **X.**
    - E.g., ***X*** is a 35-year-old customer with an income of $40,000. Suppose that *H* is the hypothesis that our customer will buy a computer.
    - Then *P*(*H*|***X***) reflects the probability that customer ***X*** will buy a computer given that we know the customer's age and income.

Bayes' Theorem: Basics

- P(H) (*prior probability*): the initial probability
  - E.g., **X** will buy computer, regardless of age, income, …

- P(X): probability that sample data is observed
  - E.g., **X** is the probability that a person from our set of customers is 35 years old and earns $40,000.
- P(X|H) (likelihood): the probability of observing the sample X, given that the hypothesis holds
  - E.g., Given that **X** will buy computer, the prob. that X is 31..40, medium income

Prediction Based on Bayes' Theorem

- Given training data **X**, *posteriori probability of a hypothesis* H, P(H|**X**), follows the Bayes' theorem

$$P(H|\mathbf{X}) = \frac{P(\mathbf{X}|H)P(H)}{P(\mathbf{X})} = P(\mathbf{X}|H) \times P(H) / P(\mathbf{X})$$

- Informally, this can be viewed as

  posteriori = likelihood x prior/evidence

- Predicts **X** belongs to C_i iff the probability P(C_i|**X**) is the highest among all the P(C_k|X) for all the *k* classes
- Practical difficulty: It requires initial knowledge of many probabilities, involving significant computational cost

Classification Is to Derive the Maximum Posteriori

- Let D be a training set of tuples and their associated class labels, and each tuple is represented by an n-D attribute vector **X** = $(x_1, x_2, \ldots, x_n)$
- Suppose there are *m* classes $C_1, C_2, \ldots, C_m$.
- Classification is to derive the maximum posteriori, i.e., the maximal $P(C_i|\mathbf{X})$
- This can be derived from Bayes' theorem

$$P(C_i|\mathbf{X}) = \frac{P(\mathbf{X}|C_i)P(C_i)}{P(\mathbf{X})}$$

- Since P(X) is constant for all classes, only

$$P(C_i|\mathbf{X}) = P(\mathbf{X}|C_i)P(C_i)$$

needs to be maximized

Naïve Bayes Classifier

- A simplified assumption: attributes are conditionally independent (i.e., no dependence relation between attributes):

$$P(\mathbf{X}|C_i) = \prod_{k=1}^{n} P(x_k|C_i) = P(x_1|C_i) \times P(x_2|C_i) \times \ldots \times P(x_n|C_i)$$

- This greatly reduces the computation cost: Only counts the class distribution
- If A_k is categorical, $P(x_k|C_i)$ is the # of tuples in C_i having value x_k for A_k divided by $|C_{i, D}|$ (# of tuples of C_i in D)
- If A_k is continous-valued, $P(x_k|C_i)$ is usually computed based on Gaussian distribution with a mean μ and standard deviation σ

  and $P(x_k|C_i)$ is $$P(\mathbf{X}|C_i) = g(x_k, \mu_{C_i}, \sigma_{C_i}) \quad (1)$$

$$g(x, \mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Naïve Bayes Classifier

- For example, let ***X*** = (35, \$40,000), where *A*1 and *A*2 are the attributes *age* and *income*, respectively.
- Let the class label attribute be *buys computer*. The associated class label for ***X*** is *yes* (i.e., *buys computer = yes*).
- Let's suppose that *age* has not been discretized and therefore exists as a continuous-valued attribute.
- Suppose that from the training set, we find that customers in *D* who buy a computer are 38 ± 12 years of age.
- In other words, for attribute *age* and this class, we have $\mu = 38$ years and $\sigma = 12$.
- We can plug these quantities, along with $x_1 = 35$ for our tuple ***X*** into Equation (1) in order to estimate *P(age = 35 | buys computer = yes)*.

Naïve Bayes Classifier: Training Dataset

Class:

C1:buys_computer = ‘yes’

C2:buys_computer = ‘no’

Data to be classified:

X = (age <=30,

Income = medium,

Student = yes

Credit_rating = Fair)

age	income	student	credit_rating	_comp
<=30	high	no	fair	no
<=30	high	no	excellent	no
31…40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31…40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31…40	medium	no	excellent	yes
31…40	high	yes	fair	yes
>40	medium	no	excellent	no

Naïve Bayes Classifier: An Example

- $P(C_i)$: P(buys_computer = "yes") = 9/14 = 0.643

  P(buys_computer = "no") = 5/14= 0.357

- Compute conditional probabilities $P(X|C_i)$ for each class

  P(age = "<=30" | buys_computer = "yes") = 2/9 = 0.222

  P(age = "<= 30" | buys_computer = "no") = 3/5 = 0.6

  P(income = "medium" | buys_computer = "yes") = 4/9 = 0.444

  P(income = "medium" | buys_computer = "no") = 2/5 = 0.4

  P(student = "yes" | buys_computer = "yes) = 6/9 = 0.667

  P(student = "yes" | buys_computer = "no") = 1/5 = 0.2

  P(credit_rating = "fair" | buys_computer = "yes") = 6/9 = 0.667

  P(credit_rating = "fair" | buys_computer = "no") = 2/5 = 0.4

Naïve Bayes Classifier: An Example

- **X = (age <= 30 , income = medium, student = yes, credit_rating = fair)**
- **$P(X|C_i)$:** P(X|buys_computer = “yes”) = *P*(*age = youth* j *buys computer = yes*)

  P(*income = medium* j *buys computer = yes*)

  P(*student = yes* j *buys computer = yes*)

  P(*credit rating = fair* j *buys computer = yes*)

  =0.222 x 0.444 x 0.667 x 0.667 = 0.044

  P(X|buys_computer = “no”) = 0.6 x 0.4 x 0.2 x 0.4 = 0.019

- To find the class, *Ci*, that maximizes *P*(***X***j*Ci*)*P*(*Ci*), we compute

$P(X|C_i)*P(C_i)$:

P(X|buys_computer = “yes”) * P(buys_computer = “yes”) = 0:044 * 0:643 = 0.028

P(X|buys_computer = “no”) * P(buys_computer = “no”) = 0:019 * 0:357 =0.007

Therefore, X belongs to class (“buys_computer = yes”)

Example - 2

Outlook	Temperature	Humidity	Windy	Class
sunny	hot	high	false	N
sunny	hot	high	true	N
overcast	hot	high	false	P
rain	mild	high	false	P
rain	cool	normal	false	P
rain	cool	normal	true	N
overcast	cool	normal	true	P
sunny	mild	high	false	N
sunny	cool	normal	false	P
rain	mild	normal	false	P
sunny	mild	normal	true	P
overcast	mild	high	true	P
overcast	hot	normal	false	P
rain	mild	high	true	N

An unseen sample

X = <rain, hot, high, false>

Play-tennis example: estimating $P(x_i|C)$

Outlook	Temperature	Humidity	Windy	Class
sunny	hot	high	false	N
sunny	hot	high	true	N
overcast	hot	high	false	P
rain	mild	high	false	P
rain	cool	normal	false	P
rain	cool	normal	true	N
overcast	cool	normal	true	P
sunny	mild	high	false	N
sunny	cool	normal	false	P
rain	mild	normal	false	P
sunny	mild	normal	true	P
overcast	mild	high	true	P
overcast	hot	normal	false	P
rain	mild	high	true	N

P(p) = 9/14
P(n) = 5/14

outlook	
P(sunny|p) = 2/9	P(sunny|n) = 3/5
P(overcast|p) = 4/9	P(overcast|n) = 0
P(rain|p) = 3/9	P(rain|n) = 2/5
temperature	
P(hot|p) = 2/9	P(hot|n) = 2/5
P(mild|p) = 4/9	P(mild|n) = 2/5
P(cool|p) = 3/9	P(cool|n) = 1/5
humidity	
P(high|p) = 3/9	P(high|n) = 4/5
P(normal|p) = 6/9	P(normal|n) = 2/5
windy	
P(true|p) = 3/9	P(true|n) = 3/5
P(false|p) = 6/9	P(false|n) = 2/5

Play-tennis example: classifying X

- An unseen sample X = <rain, hot, high, false>

- P(X|p)*P(p) =
  P(rain|p) *P(hot|p) *P(high|p) *P(false|p) *P(p) =
  3/9*2/9*3/9*6/9*9/14 = 0.010582
- P(X|n) *P(n) =
  P(rain|n) *P(hot|n) *P(high|n) *P(false|n) *P(n) =
  2/5*2/5*4/5*2/5*5/14 = 0.018286

- Sample X is classified in class n (don't play)

Example for continuous-valued attribute

- **Training set**

sex	height (feet)	weight (lbs)	foot size (inches)
male	6	180	12
male	5.92	190	11
male	5.58	170	12
male	5.92	165	10
female	5	100	6
female	5.5	150	8
female	5.42	130	7
female	5.75	150	9

Example for continuous-valued attribute

- Validation set

sex	height (feet)	weight (lbs)	foot size (inches)
Target	6	130	8

$$\hat{\mu} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

$$\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(\hat{\mu} - X_i)^2$$

$$\hat{\sigma} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(\hat{\mu} - X_i)^2}$$

	Target = male		**Target = female**	
	Mean	SD	Mean	SD
height (feet)				
weight (lbs)				
foot size(inches)				

Example for continuous-valued attribute

	Target = male		**Target = female**	
	Mean	SD	Mean	SD
height (feet)	5.885	0.027175	5.4175	0.07291875
weight (lbs)	176.25	126.5625	132.5	418.75
foot size(inches)	11.25	0.6875	7.5	1.25

Example for continuous-valued attribute

- P(male) = 0.5
- P(female) = 0.5

$$g(x,\mu,\sigma)=\frac{1}{\sqrt{2\pi}\sigma}e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

- P (h = 6| male) = g(6 , μ , σ)
- P (w = 130| male) = g(130 , μ , σ)
- P (f = 8| male) = g(8 , μ , σ)
- P (h = 6| female) = g(6 , μ , σ)
- P (w = 130| female) = g(130 , μ , σ)
- P (f = 8| female) = g(8 , μ , σ)

Example for continuous-valued attribute

- Posteriori Probability

$$P(\mathbf{X}|C_i) = g(x_k, \mu_{C_i}, \sigma_{C_i})$$

- P (X | male) =

$$p(h=6|male) * p(w=130|male) * p(\mathrm{f}=8|male) * p(male)$$

- P (X | male) =

$$p(h=6|\ female) * p(w=130|\ female) * p(\mathrm{f}=8|\ female)$$

$$* p(\ female)$$

- Highest probability

Avoiding the Zero-Probability Problem

- Naïve Bayesian prediction requires each conditional prob. be **non-zero**. Otherwise, the predicted prob. will be zero

$$P(X|C_i) = \prod_{k=1}^{n} P(x_k|C_i)$$

- Ex. Suppose a dataset with 1000 tuples, income=low (0), income= medium (990), and income = high (10)
- Use **Laplacian correction** (or Laplacian estimator)
  - *Adding 1 to each case*

    Prob(income = low) = 1/1003

    Prob(income = medium) = 991/1003

    Prob(income = high) = 11/1003
  - The “corrected” prob. estimates are close to their “uncorrected” counterparts

Naïve Bayes Classifier: Comments

- Advantages
  - Easy to implement
  - Good results obtained in most of the cases
- Disadvantages
  - Assumption: class conditional independence, therefore loss of accuracy
  - Practically, dependencies exist among variables
    - E.g., hospitals: patients: Profile: age, family history, etc.
      Symptoms: fever, cough etc., Disease: lung cancer, diabetes, etc.
    - Dependencies among these cannot be modeled by Naïve Bayes Classifier